PVG's College of Engineering & Technology and G K Pate(Wani) Institute of Management, Pune-09
Department of Artificial Intelligence & Data Science
2024-25

Final Proposal for
**The Project**
On

**Developing an Advanced Doodle Recognition System for enhancing Vocabulary Acquisition in toddlers with learning disabilities**

**Group Number: 15**

**Domain:** Machine Learning

**Team Members**

| Roll No | Name | Role/Task assigned | email | mobile | sign |
|---|---|---|---|---|---|
| 8015 | Supriya Lokhande | Team Leader | 21018015@pvgcoet.ac.in | 8080976044 | |
| 8017 | Shrinivas Patil | Team Member | 21118017@pvgcoet.ac.in | 9067849221 | |
| 8047 | Raj Kanchan | Team Member | 21118047@pvgcoet.ac.in | 9021962126 | |

**Sponsorship Details**(if any) : N/A

Name of Company:

External Guide Name:

Signature:

Email ID:

| | Name of the Faculty | Signature |
|---|---|---|
| **Internal Guide** | Prof. Mrs. V.A.Kanade | |

**Type of Project:** Application

**External Mentor**(if Any) (PG Student/Alumni/other) : N/A

**TE students associated with** (if Any):(Roll No, Name) : N/A

# Application

To,
The Project Coordinator,
Department of AI&DS
PVG's COET, Pune.

Sub:- Submission of Project Synopsis.

Respected Sir,
We undersigned, students of BE AI&DS with Group ID: 15 are submitting our project Synopsis. We are bound to the decision taken by department related to our selected project title and are submitting the final synopsis for selected project. Henceforth we will not change the project group or the selected project title/topic due to any reason.

Thanking you.

# Synopsis

## 1. Developing an Advanced Doodle Recognition System for enhancing Vocabulary Acquisition in toddlers with learning disabilities

a. Project Options:
   i. Developing an Advanced Doodle Recognition System
   ii. Streamlining Meetings: Automated Minute Generation
   iii. DocAnswer: Automated Document Question Answering

b. Problem Definition: Developing an Advanced Doodle Recognition System for Enhancing Vocabulary Acquisition in Toddlers with Learning Disabilities. Toddlers with learning disabilities sometimes find it difficult to learn using typical text-based approaches, and they also find it difficult to learn things just by reading. We wish to design and develop a system to help young learners overcome these challenges and accelerate their growth.

c. Abstract: The project develops a web-based vocabulary acquisition system that addresses challenges in traditional text-based learning for toddlers and learners with disabilities. The system leverages visual aids like images and cartoons, which have been shown to improve comprehension and retention, by transforming doodles into coherent text and digital images. The system uses an intuitive, doodle-based interface where users draw concepts on a screen, which are then processed and interpreted by a trained model. This model converts the doodles into images representing concepts or words, creating an engaging, playful learning environment. The project utilizes Google's Quick, Draw! dataset—containing 50 million doodles across 345 categories—to train an ensemble of Convolutional Neural Networks (CNNs) and sequential models. These models work together to analyse visual data and recognize patterns in the doodles, providing contextual understanding. Stable Diffusion 3 is integrated to generate detailed digital images based on text outputs from the doodles, further

enhancing the learning experience. Additionally, the project incorporates performance metrics such as retention rates, learning speed, and doodle recognition accuracy, to help young learners grasp vocabulary and concepts faster with the help of impressive visuals.

d. Goals and Objectives:

i. Design and Develop a Web-Based Doodle Recognition System: Create an intuitive online platform that transforms user-drawn doodles into coherent textual and digital representations to facilitate vocabulary instruction for toddlers with learning disabilities.

ii. Implement Advanced Recognition Models: Utilize Convolutional Neural Networks (CNNs) and sequential models to accurately interpret and categorize doodles, leveraging Google's Quick, Draw! dataset for robust recognition capabilities**.**

iii. Integrate Text-to-Image Transformation: Employ Stable Diffusion 3 to convert doodles into high-quality, detailed images that represent concepts or words, enhancing the visual learning experience.

iv. Progress Tracking: The system tracks the learner's performance, including accuracy and improvement over time, to provide insights into their learning progress and areas that may need additional focus

e. Scope of the project:

This project focuses on developing a web-based system that enhances learning experiences for toddlers with learning disabilities. The system will allow users to draw doodles that are interpreted and transformed into words and detailed images, creating a more engaging, visual-based learning environment. The use of advanced deep learning models (CNNs, LSTMs, and Sketch-RNN) for doodle recognition is combined with state-of-the-art image generation models like Stable Diffusion 3, which brings a high level of interactivity and engagement to the learning process. The project also involves tracking learner performance and providing feedback to help improve learning outcomes over time.

f. Keywords:

Doodle Recognition, Sketch Recognition, Deep Learning, Ensemble, Convolutional Neural Networks, Sequential Models, Long-Short Term Memory, Stable Diffusion, RNN

## 2. Software and hardware requirements:

- Hardware Requirements:

  GPU: NVIDIA A100 GPU 80GB

  Memory: 64 GB RAM, 250 GB of storage

  Operating System: Windows 7 64-bits or more

- Client-Side Hardware Requirements:

  CPU: Quad-core CPU (2.0 GHz or higher), Windows 7 or more

  RAM: 4 GB or higher

  Input: Touchscreen, Mouse

  Other: Stable Internet Connection (5 mbps or higher)

- Software Requirements:

  Programming Language: Python 3

  Libraries and Frameworks: PyTorch, Tensorflow, pandas, numpy, matplotlib, seaborn, scikit learn, transformers

  Front-end Tools: Streamlit

  Back-end Tools: Streamlit, FAST API, MySQL

  Preprocessing Tools: Pillow and OpenCV

  Deep Learning Networks: CNN, LSTM, Sketch-RNN, Stable DIffusion 3

  Dataset: Google's Quick! Draw Dataset

## 3. Introduction

a. Literature survey:

i. Mouna, L. E., Silkan, H., Tekouabou, S. C. K., Hanyf, Y., Tourad, M. C., & Nanne, M. F. (2024). A combined deep CNN-LSTM network for sketch recognition. International Journal of Computing and Digital Systems, 15(1), 633–644. https://doi.org/10.12785/ijcds/160147

The paper presents a novel approach for sketch recognition using a combined deep CNN-LSTM network. The CNN is used to extract features from the sketch, which are then passed to the LSTM network for classification. The proposed method outperforms both CNN and LSTM alone, as well as other state-of-the-art methods. The method achieves an accuracy of 95%, with precision and recall reaching 95%, while also achieving an F1 score of 94%. The models were trained using only 10 classes, which restricts the diversity of the data and may not fully represent the complexity of sketch recognition tasks. This limitation could affect the model's ability to generalize to a broader range of sketch categories in real-world applications.

ii. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). https://doi.org/10.1109/cvpr52688.2022.01042

This paper introduces Latent Diffusion Models (LDMs), a new approach for high-resolution image synthesis. LDMs leverage the power of diffusion models while operating in a latent space, significantly reducing computational costs compared to traditional methods. This allows for impressive image generation, text-to-image synthesis, and inpainting tasks. While LDMs achieve high fidelity and outperform previous methods in many areas, a potential drawback is the

complexity of the architecture, which might require further simplification for wider accessibility. Additionally, LDMs might struggle with generating images that contain intricate details or highly specific concepts.

iii. Kristine Guo, James WoMa, Eric Xu, "Quick, Draw! Doodle Recognition" Stanford University, 2018

The paper discusses the development of a multi-class classifier for doodles from Google's Quick, Draw! game, which involves recognizing and categorizing hand-drawn objects. The evaluation of the models employs Mean Average Precision@3 (MAP@3) to account for the high number of categories and the potential for similar objects. This metric assesses the top three predictions for each drawing, providing a more lenient evaluation compared to raw accuracy.

The results indicated that while KNN++ improves upon 1-CC, both methods struggle with categories that have vague representations. The CNN model achieved the best MAP@3 score with a specific architecture and training regimen, though it shows signs of overfitting after several epochs. The model's ability to recognize local features is highlighted, with specific examples illustrating its strengths and weaknesses in classification. Despite some misclassifications, the CNN demonstrates a superior understanding of doodle features compared to KNN models.

The authors only utilized approximately 1% of the total Quick, Draw! dataset for training their models. This limited usage may restrict the models' ability to generalize well across the diverse doodles present in the full dataset, potentially leading to lower accuracy in real-world applications. Future works suggest experimenting with more sophisticated convolutional neural network (CNN) architectures, such as VGG-Net and ResNet. These architectures have achieved state-of-the-art performance in image classification tasks, although they have not

been specifically applied to sketch recognition yet. This exploration could potentially enhance the accuracy of doodle classification.

iv. Fernandez-Fernandez, R., Victores, J. G., Estevez, D., & Balaguer, C. (2019). Quick, Stat!: a statistical analysis of the Quick, Draw! Dataset [Dataset]. In Zenodo (CERN European Organization for Nuclear Research). https://doi.org/10.5281/zenodo.7277760

The paper discusses the evolution of Deep Learning architectures, highlighting the trend of increasing layer counts to enhance performance, which necessitates large training datasets. This paper by Fernandez-Fernandez et al. (2019) analyzes the "Quick, Draw!" dataset, a massive collection of user-drawn sketches. The analysis reveals the dataset's strengths, such as its vast size and variety, making it valuable for training machine learning models in sketch recognition. The paper provides valuable insights into the Quick, Draw! Dataset, emphasizing its unique structure and the richness of the data it contains. Each sketch is stored in the ".ndjson" format, which includes essential information such as a unique identifier, category, recognition flag, timestamp, and the country of the player, allowing for comprehensive data analysis. The drawing information is represented as time series data, capturing the pencil's x and y positions along with timestamps, which facilitates a detailed examination of sketch characteristics and dynamics . Furthermore, the paper discusses a simplified version of the dataset that features aligned and scaled sketches, making it easier to process and analyze. This version is specifically utilized in training the classification Neural Network, highlighting the dataset's adaptability for machine learning applications.

v. Zhang, L. (2021). Hand-drawn sketch recognition with a double-channel convolutional neural network. *EURASIP Journal on Advances in Signal Processing*, *2021*(1), 73.

The paper presents a novel approach to hand-drawn sketch recognition, addressing the limitations of traditional deep learning methods, which often struggle with insufficient feature extraction and low recognition rates. To overcome these challenges, the author proposes a dual-channel convolutional neural network (CNN) algorithm. The process begins with preprocessing the sketch to create a smooth version, followed by applying a contour extraction algorithm to obtain the sketch's outline. Both the original sketch and its contour are then used as input for the CNN. The method incorporates feature fusion in the fully connected layer, ultimately leading to classification results generated by a softmax classifier. Experimental results demonstrate that this new approach significantly enhances the recognition rate of hand-drawn sketches, indicating its effectiveness compared to existing methods. The results show that the proposed algorithm, namely a double-channel CNN, can help improve the recognition rate of hand-drawn sketches. The authors mention that the labeled dataset used to support their findings is available from the corresponding author upon request. This indicates that while the dataset is not publicly accessible. The proposed method is primarily tested on a specific dataset, which raises concerns about its generalizability to other types of sketches or different domains. A small dataset may not adequately represent the broader population of hand-drawn sketches, leading to challenges when applying the model in varied contexts.

vi. Hayat, S., She, K., Yu, Y., & Mateen, M. (2019). Deep cnn-based features for hand-drawn sketch recognition via transfer learning approach. *Editorial Preface From the Desk of Managing Editor*, *10*(9).

The paper presents a deep convolutional neural network (DCNN) framework aimed at improving hand-drawn sketch recognition through a transfer learning approach. The proposed framework integrates three well-known pre-trained DCNN architectures: VGGNet, ResNet, and Inception-v3. These architectures are utilized in the context of transfer learning, employing a global average pooling (GAP) strategy for feature extraction. An augmented dataset is created by combining natural images with the TU-Berlin sketch dataset, which consists of 20,000 sketches across 250 categories. This augmentation aims to enhance the performance of sketch recognition. Feature maps are extracted from the input images using the three DCNN architectures, and these feature maps are concatenated. The resulting feature vector is then processed through a softmax classifier for sketch classification.

The performance of proposed individual CNN architectures, i.e., VGGNet, ResNet, and Inception-v3 obtained 78.93%, 89.61%, and 91.89% classification accuracy, respectively. However, the proposed framework achieves better performance with 94.57% sketch classification accuracy on augmented-variants TU-Berlin sketch dataset. The paper primarily focuses on the TU-Berlin sketch dataset for evaluating the proposed deep CNN-based framework for hand-drawn sketch recognition. However, considering the characteristics of the QuickDraw dataset, it could potentially yield better results for capturing sketches drawn by non-experts, which may better reflect real-world scenarios where users have varying skill levels.

vii. Ha, D., & Eck, D. (2017). Learning to Draw with Recurrent Neural Networks.arXiv preprint arXiv:1704.03477.

Learning to Draw with Recurrent Neural Networks by David Ha and Douglas Eck introduces a novel approach to generating sketch drawings using recurrent neural networks (RNNs). The model, trained on a large dataset of human-drawn

sketches, can learn to draw continuous lines based on a sequence of input vectors. This enables the model to generate realistic and coherent sketches, demonstrating its generative capabilities. While the model shows promise in various applications, it might struggle with highly complex sketches and heavily relies on the quality of the training data.

viii. P. Xu, T. M. Hospedales, Q. Yin, Y. -Z. Song, T. Xiang and L. Wang, "Deep Learning for Free-Hand Sketch: A Survey," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 45, no. 1, pp. 285-312, 1 Jan. 2023, doi: 10.1109/TPAMI.2022.3148853.

This survey paper provides a comprehensive overview of deep learning techniques for free-hand sketch recognition. It covers various topics, including feature extraction, classification, and generation of sketches. The authors discuss the challenges and limitations of existing methods and highlight promising future research directions. While the paper offers a valuable resource for researchers in the field, it may not be suitable for those with limited knowledge of deep learning or computer vision. Additionally, the paper does not delve into the ethical implications of sketch recognition technology.

ix. Kothawade, D., Kazi, I., Mhatre, J., Nair, K., & Nitnaware, P. (2020). Recognition of Labels for Hand Drawn Images. *International Research Journal of Engineering and Technology (IRJET)*, 7(5).

The paper "Recognition of Labels for Hand Drawn Images" by Kothawade et al. (2020) focuses on identifying labels written in handwritten form. This is a crucial problem in various domains, such as document analysis, form processing, and image understanding. The paper likely proposes techniques for extracting features

from hand-drawn labels and classifying them accurately. While the research could have practical implications, it might face challenges due to data limitations and the variability of handwriting styles.

x. Rombach, R., Esser, P., Saharia, C., Kaiser, T., Heng, J., & Kipf, T. (2023). ControlNet: Conditioning Diffusion Models for Image Synthesis with Spatial Control. *arXiv preprint arXiv:2302.05522*.

This paper introduces ControlNet, a novel framework that significantly enhances the controllability of diffusion models for image synthesis. By conditioning the diffusion process on additional spatial control signals, such as depth maps, poses, or edge masks, ControlNet enables users to specify desired attributes in the generated images with unprecedented precision. ControlNet offers several advantages over traditional methods. First, it provides a more intuitive and flexible way to control the image generation process compared to text-based prompts. Second, ControlNet can generate highly realistic and detailed images, even for complex scenes or objects. However, ControlNet also has some limitations. Setting up and using ControlNet can be complex, requiring users to provide appropriate control signals and understand the underlying diffusion model.

xi. Rombach, R., Esser, P., Saharia, C., Kaiser, T., Heng, J., & Kipf, T. (2023). DreamBooth: Fine-tuning Text-to-Image Diffusion Models for Subject-Specific Image Generation. arXiv preprint arXiv:2211.11560

This paper introduces DreamBooth, a groundbreaking technique that allows users to fine-tune text-to-image diffusion models to generate images of specific subjects, even if those subjects are not present in the original training data. By using a few examples of the target subject, DreamBooth can learn its unique

characteristics and generate new images that closely resemble it. DreamBooth offers several advantages over traditional methods. First, it is highly efficient and can be applied with limited training data. Second, DreamBooth can generate a wide variety of images of the target subject, including different poses, angles, and lighting conditions. However, DreamBooth also has some limitations. The quality of the generated images can be affected by the diversity and quality of the training data.

xii. Ho, J., Saharia, C., Chan, W., Girshick, R., Li, Y., Xu, Z., & He, K. (2022). Latent Diffusion Models for Image Synthesis. arXiv preprint arXiv:2112.10755

This paper introduces Latent Diffusion Models (LDMs), a powerful and flexible framework for image synthesis. LDMs operate in a latent space, making the training process more efficient and allowing for better control over the generated images. LDMs can be used to generate a wide variety of images, including realistic faces, animals, and landscapes. LDMs offer several advantages over traditional image synthesis methods. First, LDMs can generate highly realistic and detailed images, even for complex scenes or objects. Second, LDMs are computationally efficient, making them suitable for a wide range of applications. However, LDMs also have some limitations. The architecture of LDMs can be complex to understand and implement, and their performance depends on the quality and quantity of the training data.

xiii. Chen, T., Liu, J., Xu, N., Zhang, G., Li, Y., & He, K. (2023). Improving Text-to-Image Diffusion Models with Attention-Based Prompt Embeddings. *arXiv preprint arXiv:2303.05180*.

This paper introduces attention-based prompt embeddings, a novel technique that enhances the understanding of text prompts in text-to-image diffusion models. By using attention mechanisms, these embeddings can capture the nuances and

relationships between words in the prompt, leading to more accurate and relevant image generations. Attention-based prompt embeddings offer several advantages over traditional prompt embedding methods. First, they are more flexible and can handle a wider range of prompts. Second, attention-based prompt embeddings can capture the context of the prompt, which can be important for generating images that are consistent with the intended meaning. However, attention-based prompt embeddings also have some limitations. They can add complexity to the model architecture and may require additional training data. Additionally, attention-based prompt embeddings may struggle with ambiguous or contradictory prompts, leading to unexpected or unintended results.

xiv. Song, J., Bao, W., Kumar, A., LeCun, Y., & Wang, J. (2021). Diffusion Models for Image Restoration. arXiv preprint arXiv:2107.00648

This paper introduces diffusion models for image restoration, demonstrating their effectiveness in removing various types of image degradations while preserving original content and quality. Diffusion models work by gradually adding noise to an image until it becomes completely random, and then reversing this process to restore the original image. Diffusion models offer several advantages over traditional image restoration methods. First, diffusion models can handle a wide range of image degradations, including noise, artifacts, and missing information. Second, diffusion models can generate high-quality restored images that preserve the original content and structure. However, diffusion models also have some limitations. They can be computationally expensive, especially for large images or complex degradations. Additionally, diffusion models may struggle to restore images with severe damage or missing information, especially when the degradation is beyond the scope of the training data.

b. System architecture:

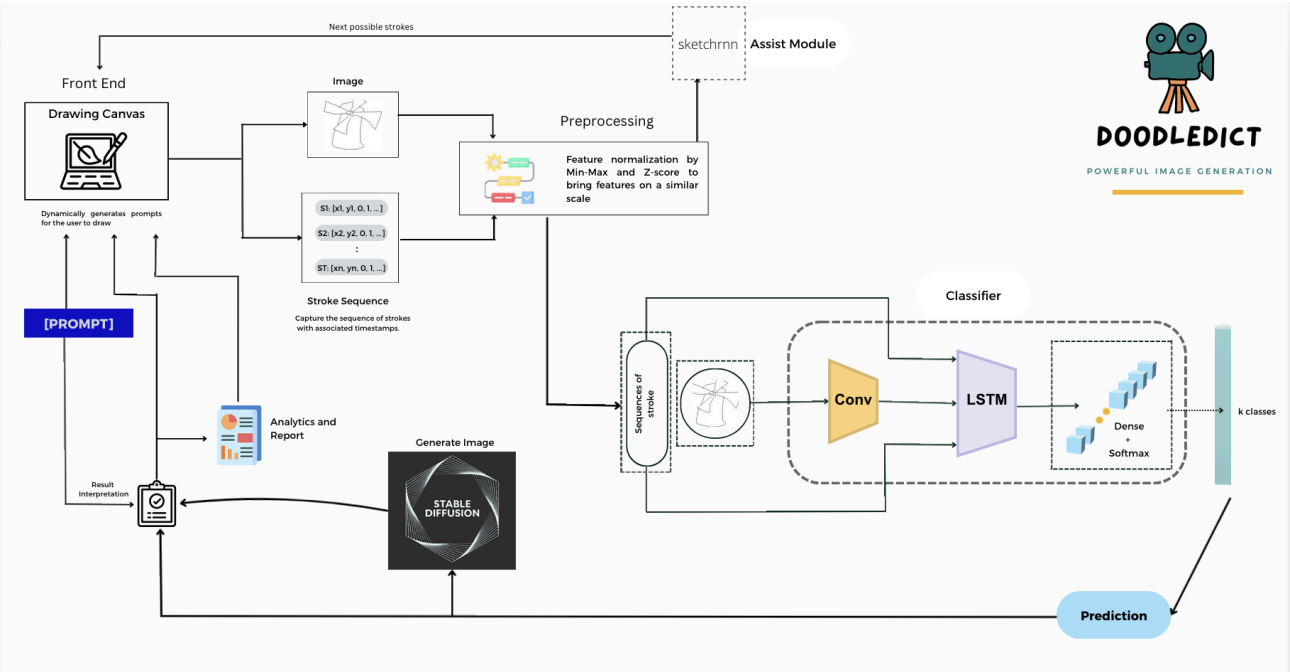
i. **Doodle Input Interface**: A touchscreen interface for toddlers to draw doodles.

ii. **Doodle Recognition Engine**: A set of CNNs and LSTMs trained on the Quick, Draw! dataset, which processes and interprets the doodles.

iii. **Text-to-Image Generator**: Stable Diffusion 3 model that converts the recognized doodles into detailed digital images.

iv. **Performance Tracker**: Module to track metrics like doodle recognition accuracy and learner progress over time.

v. **Back-end and Database**: Uses MySQL to store learner data and performance history.



c. Proposed work of project topic:-

i. Doodle Input and Interface:

- **Web-Based Drawing Canvas**: A user-friendly platform where toddlers can create doodles using a touchscreen or mouse input.
- **Real-Time Stroke Capture**: As the child draws, the system captures each stroke, storing the sequence of actions.

ii. Doodle Recognition Engine:

- **CNN and LSTM-Based Doodle Interpretation**: The system processes user-drawn doodles using Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models to analyze and interpret the strokes.
- **Sketch-RNN Integration**: Utilizes Sketch-RNN to predict and suggest the next stroke based on the user's current drawing, helping to guide them toward a complete doodle.

iii. Text-to-Image Transformation:

- **Stable Diffusion 3 Model**: Converts doodle interpretations into high-quality images, enhancing the learning experience with visually appealing content.
- **Text Output**: Displays the recognized word or concept related to the doodle.

iv. Performance Tracking and Feedback:

- **Progress Monitoring**: Tracks learner performance in terms of doodle recognition accuracy, vocabulary retention, and learning speed.
- **Feedback System**: Provides feedback and suggestions based on the accuracy of the doodles and the learner's overall progress.

v. Back-End and API Management:

- **API Endpoints**: Utilizes FastAPI to handle requests related to doodle input, recognition model predictions, image generation, and learner progress logging.

vi. Reporting and Analytics:

- **Performance Dashboard**: Displays insights and reports to educators on the learner's progress, accuracy, and areas that need improvement.

## 4. Present work related to the project topic:-

a. Current market survey:

Similar products:

i. Google Quick, Draw!

- Pros: Large dataset of doodles, fun and interactive.
- Cons: Does not focus on vocabulary acquisition or cater specifically to toddlers with learning disabilities.

ii. DoodleIt

- Pros: A tool focused on teaching CNNs for image recognition.
- Cons: Lacks integration with text-to-image transformation and progress tracking for learners.

b. Names of Conferences / Journals where papers can be published.

1. International Conference on Artificial Intelligence and Education (ICAIE)
2. International Conference on Computer Science and Education (ICCSE)
3. AI for Education Workshop at NeurIPS.
4. IEEE Frontiers in Education Conference (FIE)
5. European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)
6. International Journal of Artificial Intelligence in Education (IJAIED)
7. IEEE Transactions on Learning Technologies (TLT)
8. Education and Information Technologies (EAIT)

9. International Journal of Educational Technology in Higher Education (ETHE)
10. Journal of Learning Analytics (JLA)

c. Expected Outcome of project:

i. A fully functional web-based system that allows toddlers to draw doodles and converts them into text and images.

ii. An intuitive and interactive learning environment designed to improve vocabulary retention for toddlers with learning disabilities.

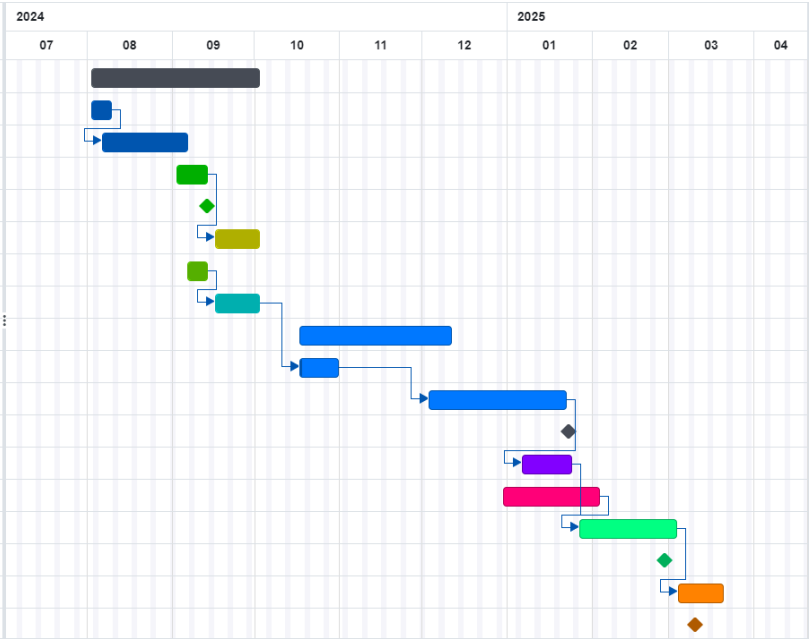
iii. A system that tracks learner performance, helping to identify areas for improvement.

d. Plan of Project Execution:-

i. Project Plan

- Requirement Analysis: Identify core functionalities, analyze datasets and model requirements.
- System Design: Design system architecture, database schema, and user interface.
- Model Development: Develop CNN, LSTM, and Sketch-RNN models for doodle recognition, and preprocess the Quick, Draw! dataset.
- Text-to-Image Integration: Configure Stable Diffusion 3 to generate images based on doodle recognition.
- Front-End Development: Build the web-based platform for drawing, recognition, and feedback.
- Testing: Conduct integration and user testing to ensure system functionality.
- Performance Monitoring: Implement learner progress tracking and feedback.

| Task Name | Start | End |
|---|---|---|
| System Design | 2024-08-02 | 2024-10-02 |
| Problem Statement | 2024-08-02 | 2024-08-09 |
| Literature Survey | 2024-08-06 | 2024-09-06 |
| SRS | 2024-09-02 | 2024-09-13 |
| Completion of System Design | 2024-09-13 | 2024-09-13 |
| System Proposal | 2024-09-16 | 2024-10-02 |
| Data Exploration | 2024-09-06 | 2024-09-13 |
| Early Model Development | 2024-09-16 | 2024-10-02 |
| Implement Stable Diffusion 3, Sketch-... | 2024-10-17 | 2024-12-11 |
| Downloading entire dataset and prepr... | 2024-10-17 | 2024-10-31 |
| Model Development and Training | 2024-12-03 | 2025-01-22 |
| Recognition and Generation Model De... | 2025-01-23 | 2025-01-23 |
| API development | 2025-01-06 | 2025-01-24 |
| Front-end development | 2024-12-30 | 2025-02-03 |
| Integrating features and APIs | 2025-01-27 | 2025-03-03 |
| Web Application Development | 2025-02-27 | 2025-02-27 |
| Testing And Deployment | 2025-03-04 | 2025-03-20 |
| Deployment | 2025-03-10 | 2025-03-10 |




ii. Probable date of completion:

24 February 2025

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